E1 Notes: differential equations and dynamical systems

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A *dynamical system* is a rule that tells you how something changes. If time jumps in steps (once a day, once a generation) the rule is a **recurrence relation** or **difference equation**. If time flows continuously the rule is a **differential equation**. This module studies both, and shows how they are related: Euler's method at the end turns a differential equation back into a difference equation.

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1. Sequences

A **sequence** (*rij*) is a list $u_0, u_1, u_2, \ldots$ of numbers. Formally it is a function $u : \mathbb{N} \to \mathbb{R}$ with u_n written for $u(n)$. Dutch textbooks usually start at $n = 0$; check whether a problem starts at 0 or 1.

There are two ways to describe a sequence:

- a **direct formula** (*directe formule*): $u_n = 3 \cdot 2^n$;
- a **recursive formula** (*recursieve formule*) plus a starting value: $u_{n+1} = 2u_n$, $u_0 = 3$.

A recursive formula is a *local* rule (how to get the next term), a direct formula is a *global* description. Much of this module is about turning the first into the second.

Arithmetic sequence (*rekenkundige rij*): $u_{n+1} = u_n + v$, so $u_n = u_0 + nv$.

Geometric sequence (*meetkundige rij*): $u_{n+1} = r\,u_n$, so $u_n = u_0 r^n$.

Sums. For an arithmetic sequence, write the sum forwards and backwards and add:

$$S = u_0 + u_1 + \cdots + u_{n-1} = \tfrac{n}{2}(u_0 + u_{n-1}).$$

For a geometric sequence with $r \neq 1$, multiply by r and subtract:

$$S = u_0 + u_0 r + \cdots + u_0 r^{n-1}, \qquad rS - S = u_0 r^n - u_0, \qquad S = u_0 \frac{r^n - 1}{r - 1}.$$

If $|r| < 1$ then $r^n \to 0$ and the infinite sum converges to $\dfrac{u_0}{1 - r}$.

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2. Linear difference equations

The most important recurrence in this module is

$$u_{n+1} = a\,u_n + b \qquad (a \neq 1).$$

Examples: a bank account with interest and a fixed monthly deposit; a medicine of which 30% is broken down each day and a fixed dose is added; a fish pond from which a fixed number of fish is taken each year.

Equilibrium value. A number L is an **equilibrium value** or **fixed point** (*evenwichtswaarde*, *dekpunt*) if starting at L you stay at L: $L = aL + b$, so

$$L = \frac{b}{1 - a}.$$

Theorem. The solution of $u_{n+1} = au_n + b$ with starting value u_0 is

$$u_n = L + a^n(u_0 - L).$$

Proof. Put $d_n = u_n - L$, the distance from equilibrium. Subtract $L = aL + b$ from $u_{n+1} = au_n + b$:

$$u_{n+1} - L = a(u_n - L), \quad \text{that is} \quad d_{n+1} = a\, d_n.$$

So (d_n) is geometric with ratio a: $d_n = a^n d_0$. Add L back. ■

This proof contains the main idea of the whole module: **measure distance from the equilibrium**, and the problem becomes simpler.

Behaviour. From $u_n = L + a^n(u_0 - L)$ you can read everything:

a	Behaviour of u_n (when $u_0 \neq L$)
$0 < a < 1$	converges to L monotonically (*monotoon*)
$-1 < a < 0$	converges to L oscillating (*slingerend*), alternately above and below
$a > 1$	moves away from L monotonically, to $\pm\infty$
$a < -1$	moves away from L oscillating, with growing amplitude
$a = -1$	jumps between two values forever (a 2-cycle)

We call L **stable** (*stabiel*, *aantrekkend*) if $|a| < 1$ and **unstable** (*instabiel*, *afstotend*) if $|a| > 1$.

Worked example. A patient takes 50 mg of a drug each morning; overnight 30% is broken down. Let u_n be the amount just after the dose on day n, with $u_0 = 50$. Then $u_{n+1} = 0.7u_n + 50$. Equilibrium: $L = 50/0.3 = 166\frac{2}{3}$ mg. Direct formula:

$$u_n = 166\tfrac{2}{3} - 116\tfrac{2}{3} \cdot 0.7^n.$$

The level rises monotonically towards $166\frac{2}{3}$ mg but never reaches it.

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3. Web diagrams

A **web diagram** (*webgrafiek*, also called a cobweb plot) shows the recurrence $u_{n+1} = f(u_n)$ geometrically.

1. Draw the graph of $y = f(x)$ and the line $y = x$.
2. Start at $(u_0, 0)$ on the x-axis. Go vertically to the graph: you are at (u_0, u_1).
3. Go horizontally to the line $y = x$: you are at (u_1, u_1).
4. Vertically to the graph: (u_1, u_2). Horizontally to the line: (u_2, u_2). Repeat.

Fixed points are exactly the intersections of the graph with $y = x$, because there $f(L) = L$.

What you see near a fixed point depends on the slope of the graph there:

- slope between 0 and 1: a **staircase** moving towards L (monotonic convergence);
- slope between -1 and 0: a **spiral** (the "web") closing in on L (oscillating convergence);
- slope greater than 1 or less than -1: the staircase or spiral moves away.

Draw a few by hand. Then make one in GeoGebra or Python: a picture with 50 steps tells you more than a table of numbers.

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4. Nonlinear recurrences and stability

Now let $u_{n+1} = f(u_n)$ with f any differentiable function. A fixed point satisfies $f(L) = L$. Near L the graph of f looks like its tangent line $y = L + f'(L)(x - L)$, so near L the recurrence behaves like a linear one with $a = f'(L)$. This suggests:

Theorem (stability criterion). Let f be differentiable with continuous derivative, and $f(L) = L$.

- If $|f'(L)| < 1$ then L is **attracting**: there is an interval around L such that for every u_0 in it, $u_n \to L$.
- If $|f'(L)| > 1$ then L is **repelling**: points close to L (but not equal to it) are pushed further away.

Proof of the first part. Because f' is continuous and $|f'(L)| < 1$, we can choose a number k with $|f'(L)| < k < 1$ and an interval $I = [L - \delta, L + \delta]$ on which $|f'(x)| \le k$. Take $u_0 \in I$. By the mean value theorem (*middelwaardestelling*) there is a ξ between u_0 and L with

$$f(u_0) - f(L) = f'(\xi)(u_0 - L).$$

Since $f(L) = L$ and $f(u_0) = u_1$ this says $|u_1 - L| = |f'(\xi)| \cdot |u_0 - L| \le k|u_0 - L|$. In particular u_1 is again in I, so we can repeat: $|u_n - L| \le k^n |u_0 - L|$. Since $0 \le k < 1$, $k^n \to 0$, so $u_n \to L$. ■

The case $|f'(L)| = 1$ is undecided: anything can happen (see "going further").

Worked example: the logistic map. Let $f(x) = rx(1 - x)$ with $r > 0$ (a population model with generations). Fixed points: $rx(1 - x) = x$ gives $x = 0$ or $1 - x = 1/r$, so $L = 1 - \frac{1}{r}$. Also $f'(x) = r - 2rx$, so

- $f'(0) = r$: the fixed point 0 is stable for $r < 1$ (the population dies out) and unstable for $r > 1$;
- $f'(1 - \frac{1}{r}) = r - 2r + 2 = 2 - r$: stable when $|2 - r| < 1$, that is $1 < r < 3$.

For $1 < r < 2$ the derivative is positive (monotonic convergence); for $2 < r < 3$ it is negative (oscillating convergence). At $r = 3$ the equilibrium loses its stability and a 2-cycle appears (problem 22). Increasing r further gives cycles of length 4, 8, 16, ... and from about $r \approx 3.57$ chaos. Run it in Python and see.

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5. Second-order linear recurrences (extension)

The Fibonacci sequence $F_{n+2} = F_{n+1} + F_n$, $F_0 = 0$, $F_1 = 1$, needs *two* previous terms. Try a geometric solution $F_n = x^n$: then $x^{n+2} = x^{n+1} + x^n$, so $x^2 = x + 1$, the **characteristic equation**. Its roots are $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$. Because the recurrence is linear, every combination $A\varphi^n + B\psi^n$ is also a solution; choosing A, B to fit F_0, F_1 gives **Binet's formula**

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}.$$

This is not required for the exam but is a beautiful use of the same idea, and you will meet it again in linear algebra (eigenvalues). Problem 20 works it out.

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6. Differential equations as models

A **differential equation** (*differentiaalvergelijking*, often abbreviated *dv*) is an equation that links an unknown function with its derivative(s). A **solution** (*oplossing*) is a function that makes the equation true for every t in an interval.

Model	Differential equation	In words
Exponential growth/decay	$\dfrac{dN}{dt} = kN$	rate of change proportional to the amount
Newton's law of cooling	$\dfrac{dT}{dt} = -k(T - T_{\text{room}})$	cooling rate proportional to the temperature difference
Limited growth (*begrensde groei*)	$\dfrac{dN}{dt} = k(G - N)$	rate proportional to the room left
Logistic growth	$\dfrac{dN}{dt} = rN\left(1 - \dfrac{N}{K}\right)$	like exponential when small, slows down near the carrying capacity K

Checking a solution. To check that $y = 3e^{2t} - 1$ solves $y' = 2y + 2$: compute the left side, $y' = 6e^{2t}$; compute the right side, $2(3e^{2t} - 1) + 2 = 6e^{2t}$. Equal for all t, so yes. Always compute both sides separately and compare: do not "manipulate the equation until it looks true".

Initial conditions. $y' = 2y + 2$ has infinitely many solutions $y = Ce^{2t} - 1$. An **initial condition** (*beginvoorwaarde*) such as $y(0) = 2$ picks one: $C - 1 = 2$, so $C = 3$. A first-order equation needs one initial condition; a second-order equation (with y'') needs two.

Why exponential growth has exactly the solutions Ce^{kt}. Suppose $N' = kN$. Look at $g(t) = N(t)e^{-kt}$. By the product rule,

$$g'(t) = N'(t)e^{-kt} - kN(t)e^{-kt} = (N' - kN)e^{-kt} = 0.$$

A function with derivative zero on an interval is constant, so $N(t)e^{-kt} = C$, that is $N(t) = Ce^{kt}$. This proves there are *no other* solutions, something that "guess and check" cannot do.

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7. Slope fields

For an equation $y' = f(x, y)$, the right-hand side tells you, at every point (x, y), the slope that a solution curve through that point must have. Draw at many points a short line segment with that slope: this is the **slope field** (*lijnelementenveld*, direction field). Solution curves (*oplossingskrommen*) follow the segments, like iron filings show magnetic field lines.

Isoclines. The set of points where the slope has a fixed value c is the curve $f(x, y) = c$ (an *isocline*). For $y' = x - y$ the isoclines are the lines $y = x - c$. On $y = x - 1$ the slope is 1, the same as the slope of the line itself, so the line $y = x - 1$ *is* a solution curve.

Autonomous equations and equilibria. If the right-hand side does not depend on t, $y' = g(y)$, the equation is **autonomous**. Then:

- every zero y^* of g gives a constant solution $y(t) = y^*$, an **equilibrium solution** (*evenwichtsoplossing*);
- between two equilibria, y is monotonic (the sign of g is fixed there);
- y^* is **stable** if solutions nearby move towards it: $g > 0$ just below y^* and $g < 0$ just above. A sufficient test: $g'(y^*) < 0$ means stable, $g'(y^*) > 0$ means unstable.

Compare: for recurrences the test was $|f'(L)| < 1$, for differential equations it is $g'(y^*) < 0$. They agree once you look at Euler's method (§10): one Euler step is the map $y \mapsto y + hg(y)$, with derivative $1 + hg'(y^*)$, which has absolute value less than 1 for small $h > 0$ exactly when $g'(y^*) < 0$.

For the logistic equation $g(N) = rN(1-N/K)$ the equilibria are $N = 0$ (unstable, $g'(0) = r > 0$) and $N = K$ (stable, $g'(K) = -r < 0$).

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8. Separation of variables

An equation is **separable** if it can be written as

$$\frac{dy}{dx} = g(x)\,h(y).$$

Method (*scheiden van variabelen*):

1. Find all constant solutions: every y^* with $h(y^*) = 0$ gives $y = y^*$. Write them down now; step 2 loses them.
2. Where $h(y) \neq 0$, divide: $\dfrac{1}{h(y)}\dfrac{dy}{dx} = g(x)$.
3. Integrate both sides: $\displaystyle\int \frac{1}{h(y)}\,dy = \int g(x)\,dx + C$.
4. Solve for y if possible, and use the initial condition to find C.

Why it works. Let H be an antiderivative of $1/h$ and G one of g. If $y(x)$ solves the equation, then by the chain rule

$$\frac{d}{dx}H(y(x)) = H'(y(x))\,y'(x) = \frac{y'(x)}{h(y(x))} = g(x) = G'(x).$$

So $H(y(x)) - G(x)$ has derivative zero and is constant: $H(y) = G(x) + C$. That is exactly step 3. The "multiply by dx" trick in textbooks is shorthand for this chain-rule argument.

Worked example 1. $y' = 2xy$, $y(0) = 3$. Constant solution: $y = 0$ (does not fit $y(0) = 3$). Otherwise

$$\int \frac{dy}{y} = \int 2x\,dx \;\Rightarrow\; \ln|y| = x^2 + C \;\Rightarrow\; |y| = e^C e^{x^2} \;\Rightarrow\; y = Ae^{x^2}.$$

Here $A = \pm e^C$ can be any nonzero number, and $A = 0$ gives back the constant solution. With $y(0) = 3$: $y = 3e^{x^2}$.

Worked example 2: Newton's law of cooling. $T' = -k(T - 20)$, $T(0) = 90$. Put $D = T - 20$; then $D' = -kD$, so $D = 70e^{-kt}$ and $T = 20 + 70e^{-kt}$. If after 5 minutes $T = 70$: $70e^{-5k} = 50$, so $k = \frac{1}{5}\ln\frac{7}{5} \approx 0.0673$ per minute.

Worked example 3: the logistic equation. $N' = rN(1 - N/K)$ with $0 < N_0 < K$. Separate:

$$\int \frac{K}{N(K-N)}\,dN = \int r\,dt.$$

Use partial fractions: $\dfrac{K}{N(K-N)} = \dfrac{1}{N} + \dfrac{1}{K-N}$ (check by adding). So

$$\ln N - \ln(K-N) = rt + C \;\Rightarrow\; \frac{N}{K-N} = Be^{rt}, \quad B = \frac{N_0}{K-N_0}.$$

Solve for N:

$$N(t) = \frac{K}{1 + Ae^{-rt}}, \qquad A = \frac{K-N_0}{N_0}.$$

Worked example 4: a warning. $y' = y^2$, $y(0) = 1$. Separating gives $-1/y = x + C$, $C = -1$, so $y = \dfrac{1}{1-x}$. This solution exists only for $x < 1$: it **blows up** at $x = 1$ although the equation looks harmless. A solution of a differential equation lives on an interval, and the interval matters.

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9. Exponential and logistic growth compared

	Exponential $N' = rN$	Logistic $N' = rN(1 - N/K)$
Solution	$N_0 e^{rt}$	$\dfrac{K}{1 + Ae^{-rt}}$
Long term	$\to \infty$	$\to K$
Growth rate N'	grows with N	largest at $N = K/2$
Graph	convex	S-shaped, inflection point at $N = K/2$

Why the inflection point is at $K/2$. You do not need the solution. The growth rate $N' = g(N) = rN - \frac{r}{K}N^2$ is a downward parabola in N with top at $N = K/2$, so the growth is fastest when $N = K/2$. Differentiating, $N'' = g'(N)N'$, and since $N' > 0$ for $0 < N < K$, N'' changes sign exactly where $g'(N) = r - 2rN/K$ does, at $N = K/2$. The maximal growth rate is $g(K/2) = rK/4$.

Doubling time and half-life. For $N = N_0 e^{rt}$ the doubling time (*verdubbelingstijd*) is $\ln 2/r$; for decay $N_0 e^{-kt}$ the half-life (*halveringstijd*) is $\ln 2/k$.

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10. Euler's method

Most differential equations cannot be solved with a formula. **Euler's method** (*methode van Euler*) approximates the solution of $y' = f(t, y)$, $y(t_0) = y_0$, by walking along the slope field in small steps of size h:

$$t_{k+1} = t_k + h, \qquad y_{k+1} = y_k + h\, f(t_k, y_k).$$

In words: at the current point, read off the slope, follow the tangent line for a time h. This turns the differential equation into a **difference equation**: the two halves of this module meet.

As an algorithm (wiskunde techniek, *digitaal gereedschap*):

```
def euler(f, t0, y0, h, n):
    t, y = t0, y0
    for _ in range(n):
        y = y + h * f(t, y)
        t = t + h
    return y

# example: y' = t + y, y(0) = 1, approximate y(1)
for n in (2, 10, 100, 1000):
    print(n, euler(lambda t, y: t + y, 0, 1, 1/n, n))
```

The exact solution is $y = 2e^t - t - 1$, so $y(1) = 2e - 2 \approx 3.4366$. With $n = 2$ steps Euler gives 2.5; with $n = 1000$ it gives about 3.4338.

On a graphing calculator use the sequence mode (*rijen-modus*) or a short program: the Euler recurrence is just a recursive sequence.

How large is the error? For $y' = y$, $y(0) = 1$, one step multiplies by $1 + h$, so n steps of size $h = 1/n$ give $y(1) \approx (1 + 1/n)^n$:

n	$(1 + 1/n)^n$	error $e - (1 + 1/n)^n$
10	2.59374	0.1245
100	2.70481	0.0135
1000	2.71692	0.0014

Ten times more steps gives about ten times smaller error: the global error of Euler's method is roughly proportional to h. We say Euler's method is **first-order**. (The reason: each step makes an error of about $\frac{1}{2}h^2 y''$, from the Taylor expansion, and there are $1/h$ steps.)

Euler's method also explains the definition $e = \lim_{n\to\infty}(1 + 1/n)^n$ that you met in stage C.

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11. Common mistakes

1. **Forgetting the constant, or putting it in the wrong place.** $\ln y = x^2 + C$ gives $y = Ae^{x^2}$, not $y = e^{x^2} + C$.
2. **Losing solutions when dividing.** Dividing by y throws away $y = 0$. List constant solutions first.
3. **Mixing up u_n and u_{n+1}** in a web diagram: horizontal moves go to the line $y = x$, vertical moves to the graph.
4. **Calling an unstable fixed point "the limit".** A sequence converges to L only if L is stable *and* u_0 is in its basin of attraction. Starting exactly at an unstable fixed point you stay there, but any rounding error moves you away.
5. **Using $f'(L)$ for a differential equation the way you use it for a recurrence.** For $y' = g(y)$ the test is the sign of $g'(y^*)$, not whether $|g'(y^*)| < 1$.
6. **Euler with the wrong slope.** Use the slope at the *start* of the step, $f(t_k, y_k)$, not at the end.
7. **Checking a solution by rearranging.** Compute left side and right side separately.

8. **Ignoring the interval.** $y = 1/(1-x)$ solves $y' = y^2$ only on $x < 1$ (for the initial condition $y(0) = 1$).

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12. Going further

- **Better numerical methods.** Heun's method uses the average of the slopes at the start and at the Euler-predicted end; its error is proportional to h^2. The classical Runge–Kutta method (RK4) has error proportional to h^4 and is what computers usually use. Implement both and compare errors as h halves.
- **Linear first-order equations.** $y' + p(t)y = q(t)$ is solved by multiplying by the *integrating factor* $e^{\int p\,dt}$; the left side becomes $(ye^{\int p\,dt})'$. Problem 23.
- **Systems.** Predator–prey (Lotka–Volterra): $x' = ax - bxy$, $y' = -cy + dxy$. Two populations, solution curves are closed loops in the (x, y)-plane. Euler's method works the same way with vectors.
- **Chaos.** The logistic map at $r = 4$ is chaotic: two starting values differing by 10^{-10} give completely different sequences after about 35 steps. Read about Feigenbaum's constant $\delta \approx 4.669$ in the period-doubling cascade.
- **The undecided case $|f'(L)| = 1$.** For $f(x) = \sin x$, $L = 0$ and $f'(0) = 1$, yet $u_n \to 0$ (very slowly, roughly like $\sqrt{3/n}$). For $f(x) = x + x^3$, also $f'(0) = 1$, but 0 repels. The first-derivative test cannot tell these apart.
- **University connection.** This material is the start of the first-year course *Analyse I Differentiaalvergelijkingen*. Strogatz's *Nonlinear Dynamics and Chaos* is the standard next step.